

Exp No: 14

Experiment to Understand Feature Scaling

Aim

To understand the importance of feature scaling and apply **Standard Scaling and Min-Max Scaling** to the given dataset.

Program Description

The program loads the dataset and handles missing values using mean imputation. Categorical data is converted into numerical form using One-Hot Encoding. The numerical features are then scaled using **StandardScaler** and **MinMaxScaler**.

Program:

```
import numpy as np

import pandas as pd

df=pd.DataFrame({
'Country':['France','Spain','Germany','Spain','Germany','France','Spain',np.nan,'France','France'],

    'Age':[44,27,30,38,40,35,np.nan,48,50,37],

    'Salary':[72000,48000,54000,61000,np.nan,58000,52000,79000,83000,67000],

    'Purchased':['No','Yes','No','No','Yes','Yes','No','Yes','No','Yes']
})

print(df)

print(df.head())

df.Country.fillna(df.Country.mode()[0],inplace=True)

features=df.iloc[:, :-1].values

label=df.iloc[:, -1].values

from sklearn.impute import SimpleImputer

age=SimpleImputer(strategy="mean",missing_values=np.nan)

Salary=SimpleImputer(strategy="mean",missing_values=np.nan)

age.fit(features[:,[1]])

Salary.fit(features[:,[2]])
```

```

features[:,[1]]=age.transform(features[:,[1]])
features[:,[2]]=Salary.transform(features[:,[2]])
print(features)

from sklearn.preprocessing import OneHotEncoder
oh = OneHotEncoder(sparse_output=False)
Country=oh.fit_transform(features[:,[0]])
print(Country)

final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
print(final_set)

from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)

feat_standard_scaler=sc.transform(final_set)
print(feat_standard_scaler)

from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)

feat_minmax_scaler=mms.transform(final_set)
print(feat_minmax_scaler)

```

Output:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No

7 NaN 48.0 79000.0 Yes

8 France 50.0 83000.0 No

9 France 37.0 67000.0 Yes

Country Age Salary Purchased

0 France 44.0 72000.0 No

1 Spain 27.0 48000.0 Yes

2 Germany 30.0 54000.0 No

3 Spain 38.0 61000.0 No

4 Germany 40.0 NaN Yes

['France' 44.0 72000.0]

['Spain' 27.0 48000.0]

['Germany' 30.0 54000.0]

['Spain' 38.0 61000.0]

['Germany' 40.0 63777.77777777778]

['France' 35.0 58000.0]

['Spain' 38.77777777777778 52000.0]

['France' 48.0 79000.0]

['France' 50.0 83000.0]

['France' 37.0 67000.0]]

[[1. 0. 0.]

[0. 0. 1.]

[0. 1. 0.]

[0. 0. 1.]

[0. 1. 0.]

[1. 0. 0.]

[0. 0. 1.]

[1. 0. 0.]

[1. 0. 0.]

[1. 0. 0.]]
[[1.0 0.0 0.0 44.0 72000.0]
[0.0 0.0 1.0 27.0 48000.0]
[0.0 1.0 0.0 30.0 54000.0]
[0.0 0.0 1.0 38.0 61000.0]
[0.0 1.0 0.0 40.0 63777.77777777778]
[1.0 0.0 0.0 35.0 58000.0]
[0.0 0.0 1.0 38.77777777777778 52000.0]
[1.0 0.0 0.0 48.0 79000.0]
[1.0 0.0 0.0 50.0 83000.0]
[1.0 0.0 0.0 37.0 67000.0]]
[[1.00000000e+00 -5.00000000e-01 -6.54653671e-01 7.58874362e-01
7.49473254e-01]
[-1.00000000e+00 -5.00000000e-01 1.52752523e+00 -1.71150388e+00
-1.43817841e+00]
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-5.26656882e-01]
[-1.00000000e+00 -5.00000000e-01 1.52752523e+00 0.00000000e+00
-1.07356980e+00]
[1.00000000e+00 -5.00000000e-01 -6.54653671e-01 1.34013983e+00
1.38753832e+00]
[1.00000000e+00 -5.00000000e-01 -6.54653671e-01 1.63077256e+00

```

1.75214693e+00]
[ 1.00000000e+00 -5.00000000e-01 -6.54653671e-01 -2.58340208e-01
 2.93712492e-01]]
[[2.    0.    0.    1.47826087 1.37142857]
 [0.    0.    2.    0.    0.    ]
 [0.    2.    0.    0.26086957 0.34285714]
 [0.    0.    2.    0.95652174 0.74285714]
 [0.    2.    0.    1.13043478 0.9015873 ]
 [2.    0.    0.    0.69565217 0.57142857]
 [0.    0.    2.    1.02415459 0.22857143]
 [2.    0.    0.    1.82608696 1.77142857]
 [2.    0.    0.    2.    2.    ]
 [2.    0.    0.    0.86956522 1.08571429]]

```

/tmp/ipykernel_608/3439832340.py:15: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
```

Result:

Thus, the given dataset was successfully preprocessed and the features were scaled using StandardScaler and MinMaxScaler.